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SDGs Alignment of Indonesia’s Research Strength

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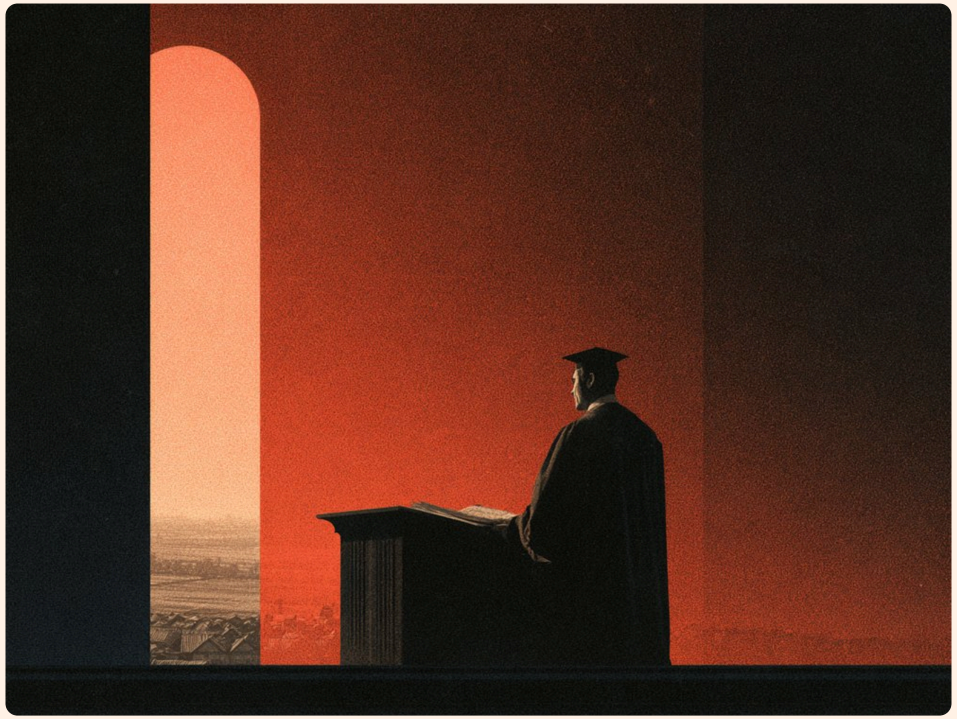
▸ BIO

Indonesia’s universities publish a great deal on the SDGs. Three questions follow: whether that output goes where provinces need it, whether counting it says anything about its quality, and who actually produces it.



I Does research go where it is needed?

Each province has goals it is failing and goals its universities are strong in.
They are rarely the same goals.

METHOD

How alignment was measured

Supply and demand are ranked over the same 16 goals (SDGs 1–16) for each of the 34 provinces, then compared.

1

Supply

Scopus publication counts per SDG for **457** Indonesian universities, attributed to the province the university sits in. Each province's **five strongest goals** are its supply.

2

Measured demand

285 official indicator-dimensions from the National SDGs Secretariat, aggregated by principal component analysis into a 0–100 achievement index. Each province's **five weakest goals** are its demand.

3

Stated demand

A survey of provincial Bappeda, February–March 2025, answered by **27 of 34 provinces** — their own declared priorities. Complemented by focus groups and content analysis of all 34 provincial RPJMDs.

4

Alignment

A goal on both lists is a match — a development gap the province's own universities are already strong in.

FIGURE 1

Where universities are strong, where provinces are failing, and what provinces say they want

All 16 goals, ordered by the gap between revealed demand and supply. Demand is measured twice, from two different sources.

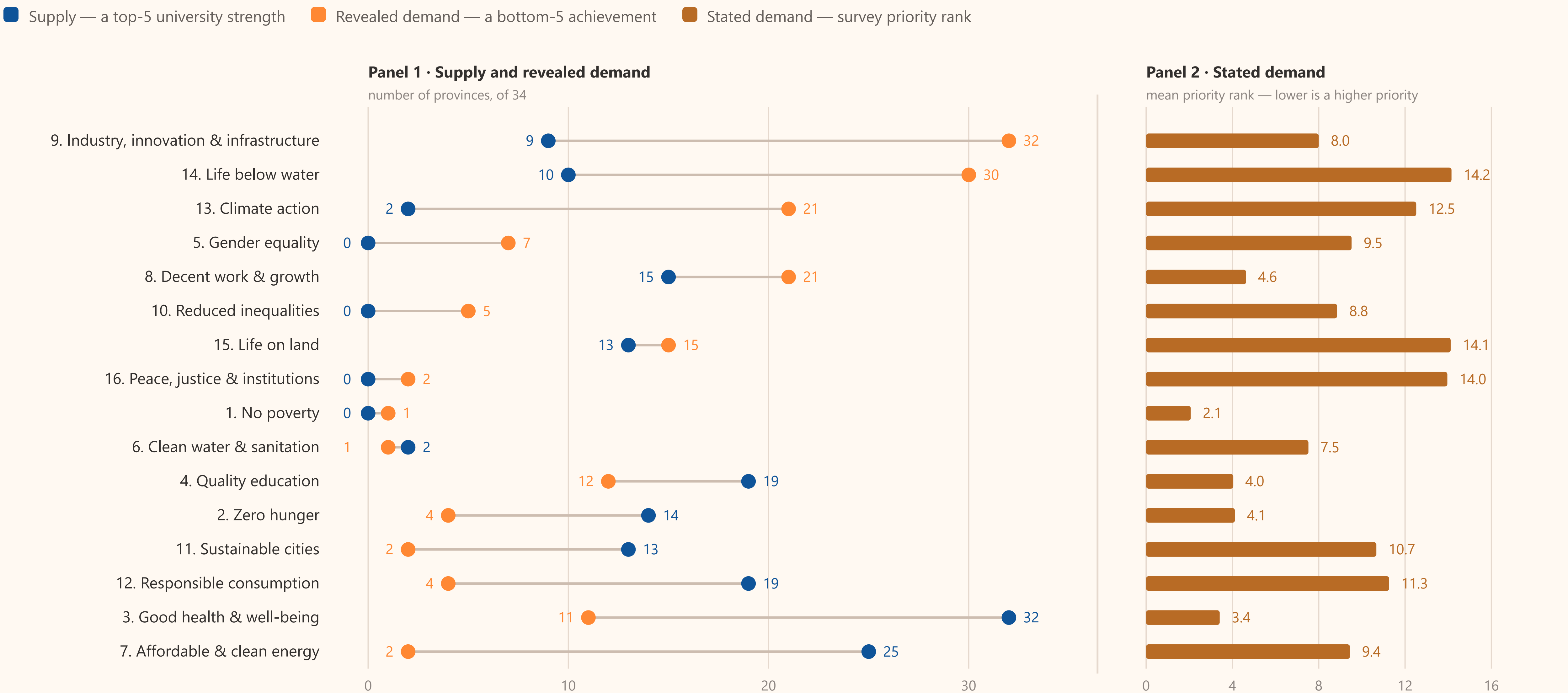
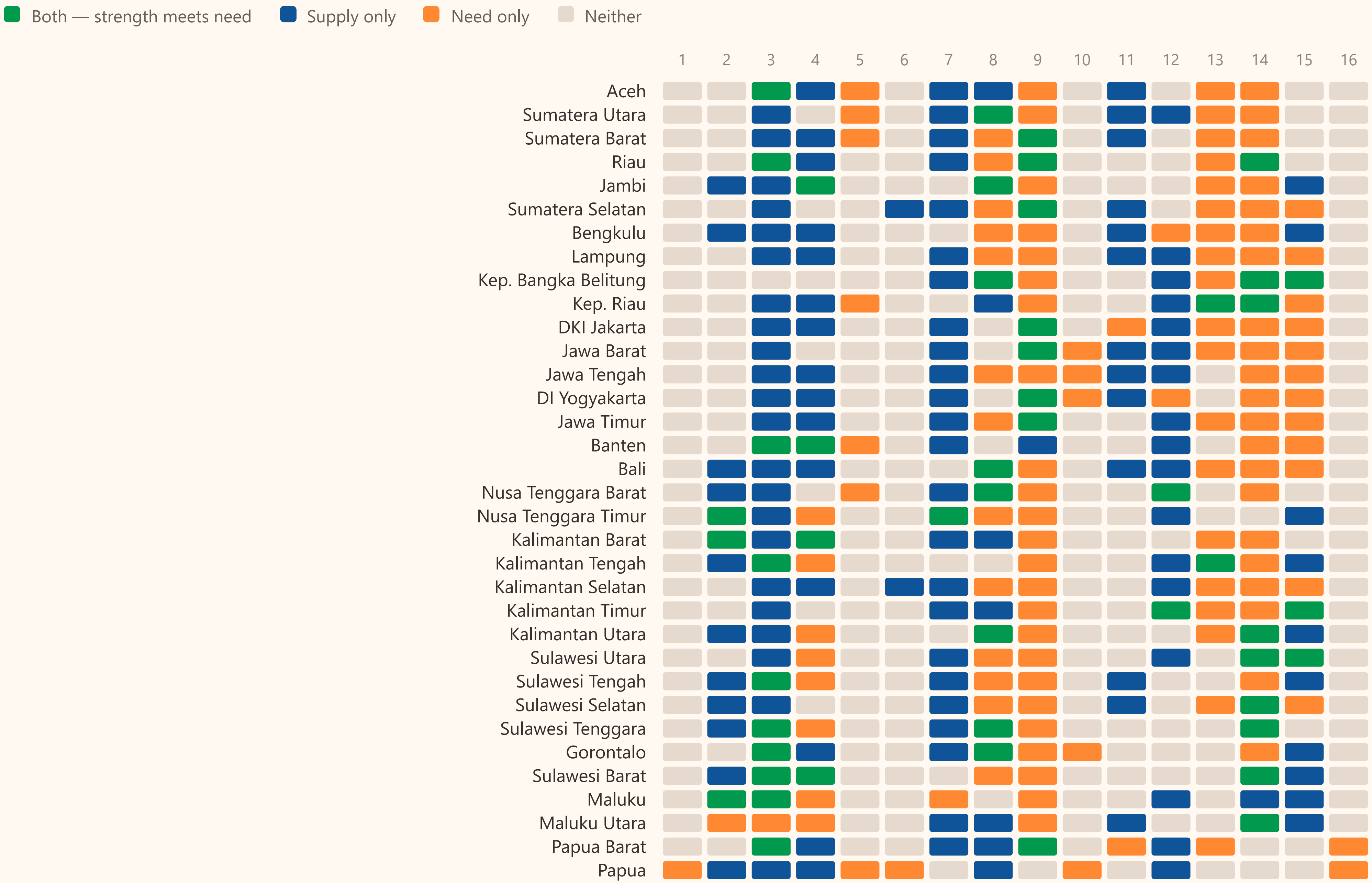


FIGURE 2

Province by province, goal by goal

Each row is a province, each column an SDG. Green is a match: a goal that is both one of the province's five strongest by publication and one of its five weakest by achievement.



II Does counting research tell you anything?

Rank the same institutions twice — once by how much they publish, once by how much of it reaches a Q1 journal — and the two answers disagree.

METHOD

How research quality was measured

978,546 journal articles and reviews from Indonesia's 30 largest research producers, 2015–2026, indexed on OpenAlex with no topic filter and cross-classified against Scopus.

1

Journal quality

Every source matched by ISSN against the Scopus CiteScore quartile list, May 2026 — Q1 to Q4, delisted, indexed-but-unranked, or not indexed at all.

2

Coverage check

OpenAlex's journal register was tested against Elsevier's own API: 250 randomly drawn “not in Scopus” journals returned **zero** false negatives. OpenAlex is a reliable census of the literature.

3

Institution check

OpenAlex is less reliable on *who wrote it*. **Three** of the 25 largest institution records — sharing name templates like *IAIN X* — conflate a whole class of institutions: established universities matched their own affiliation strings at 92–99%, these three at 1–6%. All three are excluded.

4

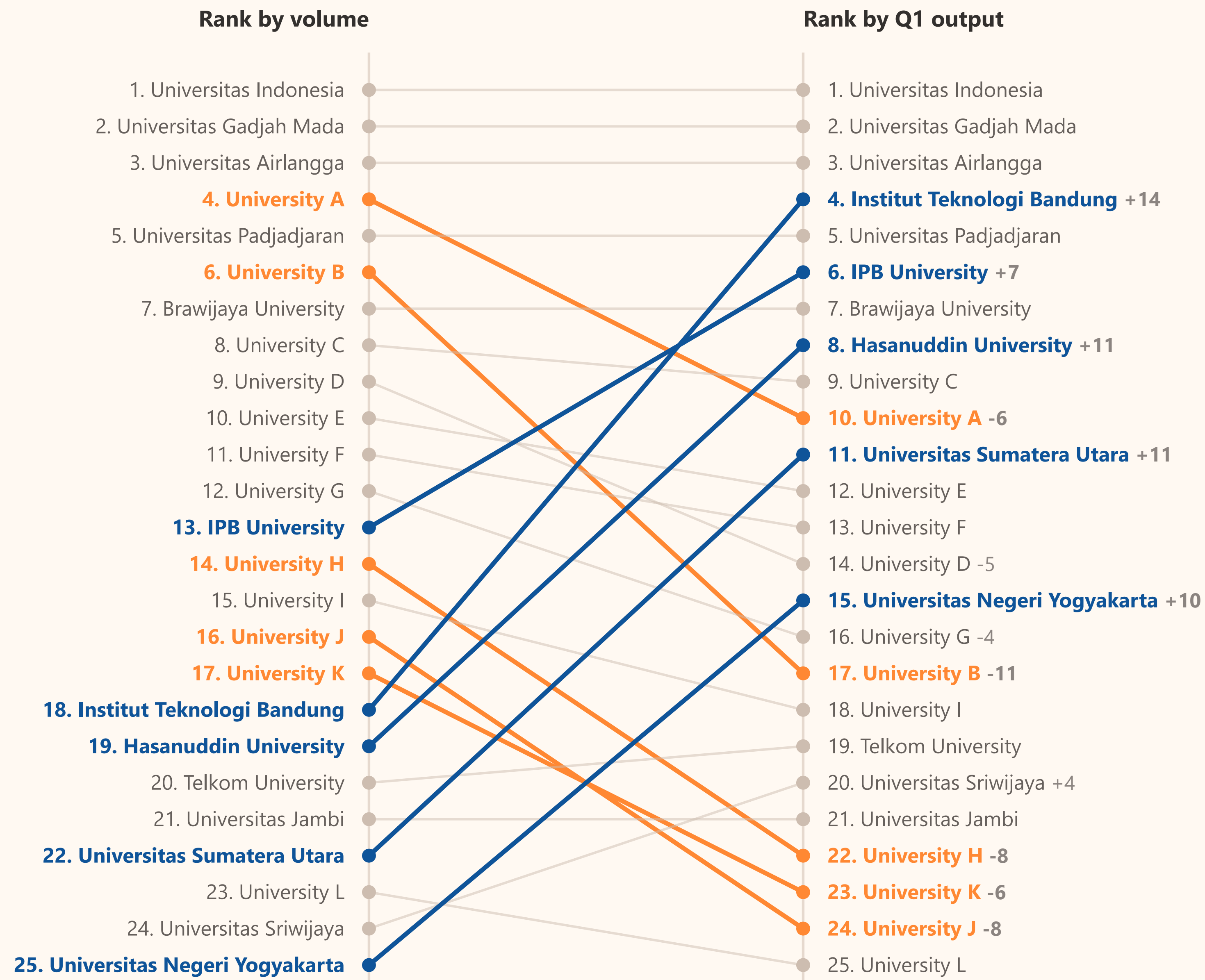
Ranking

The 25 largest remaining producers, ranked twice over the same period: once by total output, once by Q1-only output.

FIGURE 3

Counting everything inverts the ranking

Only 5.2% of the 978,546 works reach a Q1 journal. Institutions that gain ground under the quality filter are named; those that lose ground are shown as letters, held consistent across both columns.



III What the published dashboard shows

The SDG Center's dashboard maps Indonesia's Q1 output onto the seventeen goals. The landscape it describes is uneven and heavily concentrated.

METHOD

How the dashboard measures research strength

SDG Center UNPAD's public dashboard maps Indonesia's highest-quality research output onto the 17 goals. Live at sdgcenter.unpad.ac.id/dashboard since 18 June 2026.

1

Corpus

75,892 Q1 journal articles with an Indonesian affiliation, 2015 onwards, downloaded from the Scopus API. Articles only; discontinued titles excluded.

2

Mapping to goals

Matched to goals by the **Elsevier Aurora** keyword framework. **32,257** papers carry a tag; one paper can match several goals, so shares exceed 100%.

3

Classing the goals

Each goal's share is classed **Strong** (≥5%), **Moderate** (1–5%) or **Gap** (<1%) — the dashboard's own threshold.

4

Institutions

Co-authorship frequency: a paper counts once per institution. Universities only; agencies and hospitals excluded.

5

Concentration

Gini coefficient and Lorenz curve across all **2,571** universities in the corpus, computed from the underlying paper records.

WHAT THE METHOD CANNOT DO

Keyword tagging mis-assigns. In testing, 25.7% of the output tagged to SDG 9 was education research caught by the word *innovation*.

The Q1 filter is not neutral. Equity and urban research skews to national journals, which it excludes by construction.

Quartiles are field-normalised. Q1 in a small category is a lower bar than Q1 in medicine.

FIGURE 4

Five goals carry the output; four are near-absent

Share of Indonesia's 75,892 Q1 papers addressing each SDG, with paper counts. Four goals fall below the 1% line the dashboard uses to flag a research gap.

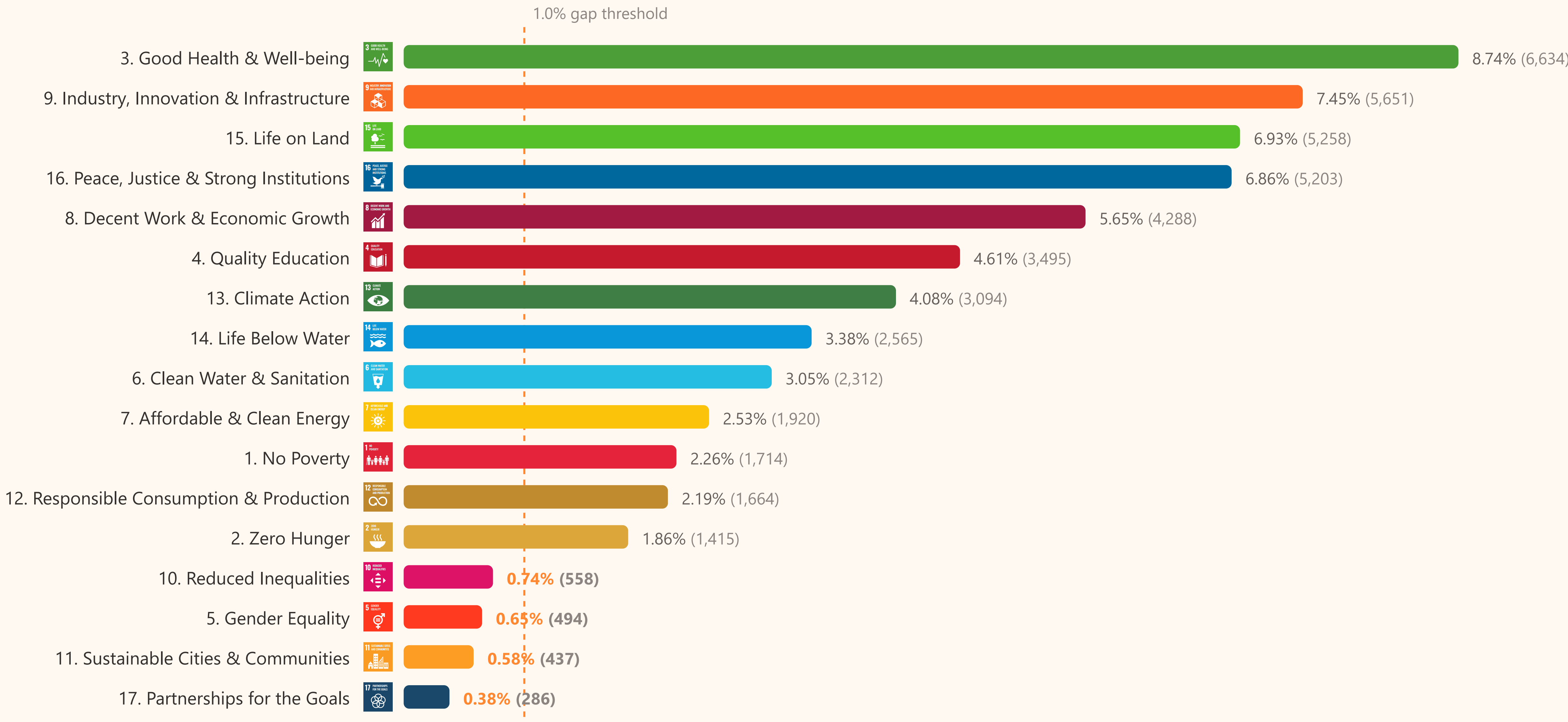


FIGURE 5

Which university leads a goal tracks what that university is for

Rank of each producer within each SDG. ITB's block sits on water, energy and consumption; IPB's on climate, marine and land; UI and Unair hold health. The two teacher-training universities barely register outside Quality Education, which they lead.

ranked 1st in the goal · 2nd–3rd · 4th–5th · 6th–10th · outside the top 10

	UI	UGM	Unair	Unpad	ITB	IPB	Brawijaya	Hasanuddin	Undip	UNS	UM Malang	UPI
1. No Poverty	1	2	6	5	9	3	7	4	10	8		
2. Zero Hunger	1	3	5	2	10	4	6	7	8	9		
3. Good Health & Well-being	1	3	2	4		10	6	5	9			
4. Quality Education	2	6	5	9						7	1	3
5. Gender Equality	1	2	3	7								
6. Clean Water & Sanitation	4	2	3	9	1	7	8	10	6			
7. Affordable & Clean Energy	2	3	6		1		10		7	5		
8. Decent Work & Economic Growth	1	2	3	4	6	5	7			9	10	
9. Industry, Innovation & Infrastructure	1	3	4	5	2		8		9	6		
10. Reduced Inequalities	1	4	2	3	5		8	7		6		
11. Sustainable Cities & Communities	1	3		5	2	6	10		7	4		
12. Responsible Consumption & Production	2	4	6	10	1	5	8		7	9		
13. Climate Action	4	2	6	7	3	1	10	8	5	9		
14. Life Below Water	8	5		7	2	1	9	3	4			
15. Life on Land	3	2	9	7	4	1	6	5	8			
16. Peace, Justice & Strong Institutions	2	1	3	4		8	6	7	10	5		
17. Partnerships for the Goals	2	1	10	4	9	3		5	8	6		

FIGURE 6

Productivity and breadth pick out different leaders

Publication volume against the number of the 17 goals in which a university ranks in the top ten. UI and UGM lead on both. Unpad and ITB publish almost the same amount; Unpad reaches the top ten in 16 goals, ITB in 13.

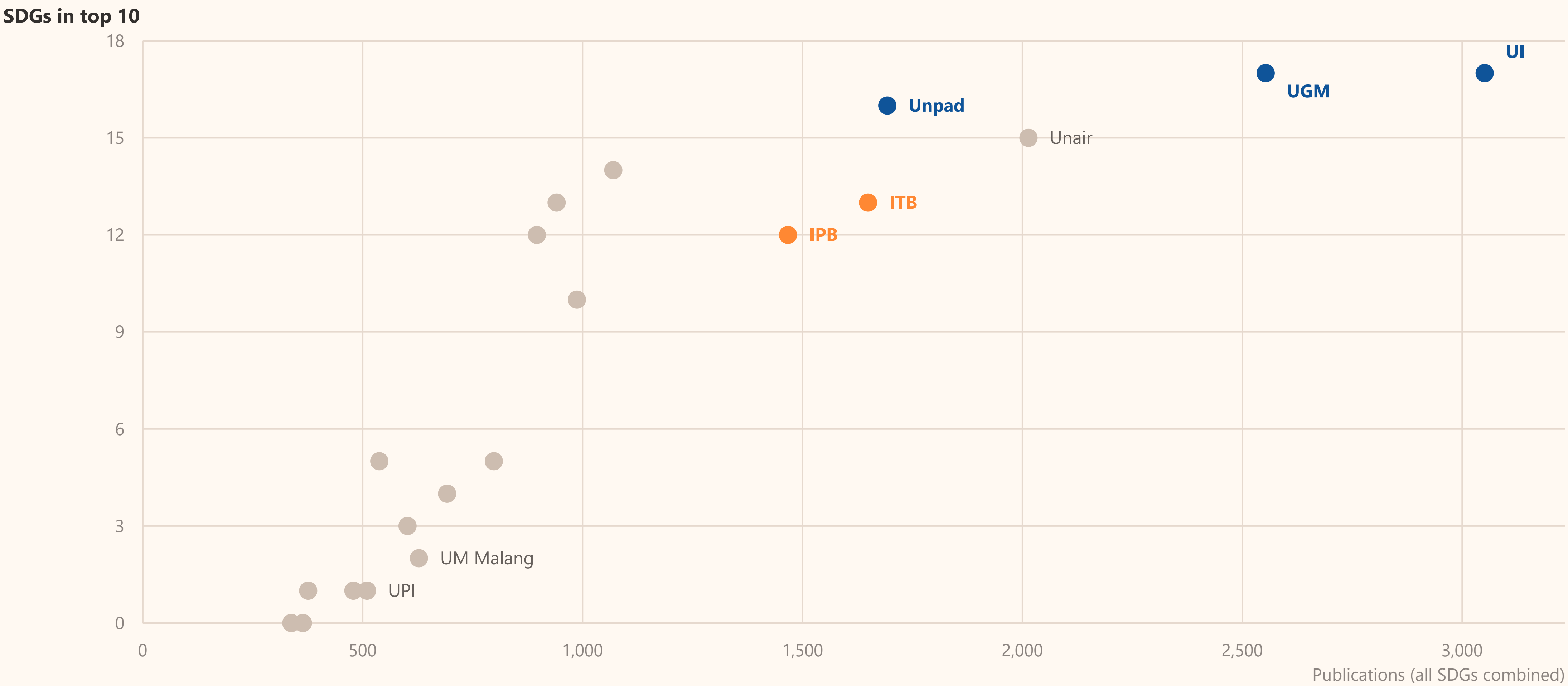
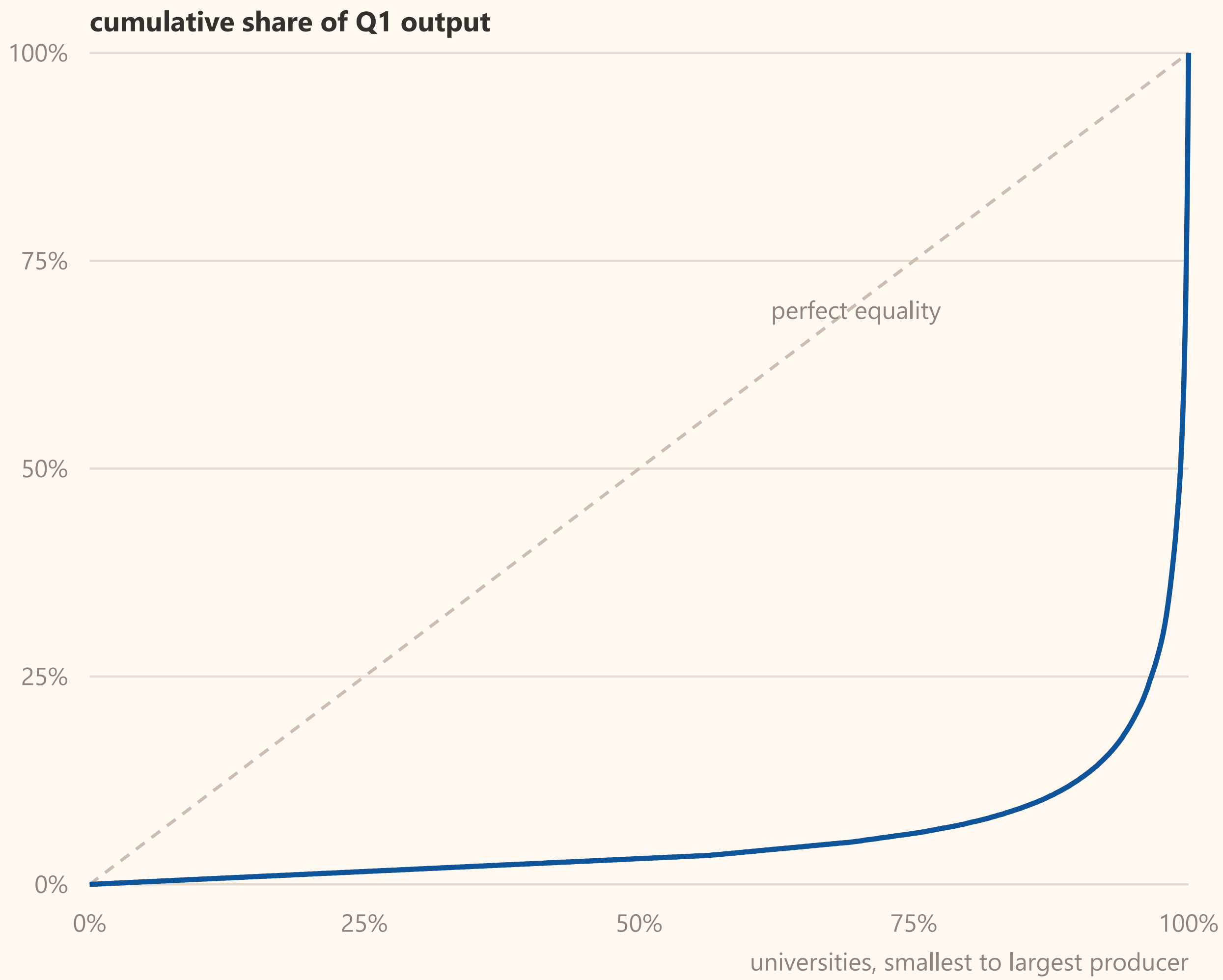


FIGURE 7

Expertise is extraordinarily concentrated

Cumulative share of Q1 SDG output across the 2,571 universities in the corpus, ordered from smallest to largest producer. The diagonal is perfect equality.



0.89

Gini coefficient across 2,571 universities

38%

of all Q1 SDG output held by the top ten institutions

1,450

universities — 56% of those in the corpus — appear on exactly one paper in eleven years

IMPLICATIONS

What follows for policy

What the evidence implies for how research capacity is measured, funded and distributed.

1 Stop ranking on volume

Counting everything inverts the ranking — ITB moves **18→4** once quality is applied. Rules keyed to raw counts reward the wrong institutions and feed the 78.8% of output no index carries. Report volume and quality separately, never merged.

FIGURE 3

2 Commission the blind spots

Gender, inequality, cities and partnerships each sit **below 1%**, and no institution has claimed any as a niche. That is a gap in incentives, not capability — it needs earmarked calls, not general capacity funds.

FIGURE 4

3 Match goals to mandates

Strength follows institutional purpose: ITB on energy, water and cities; IPB on land, climate and marine; the LPTK on education. Commission the institution whose mandate fits the goal, rather than asking all 17 goals of every university.

FIGURE 5 · REPORT REC. 4

4 Academic mobility and professor exchange

Ten institutions hold **38%** of frontier output while **1,450** universities appear on a single paper. A national visiting-professor and sabbatical scheme moves capability to where it is thin faster than building it from nothing.

FIGURE 7

5 A national consortium of SDG Centers

SDG 17 is the weakest goal at **0.38%** and simultaneously the instrument for the rest. Inter-provincial collaboration, joint research agendas and consortium authorship are how the tail reaches the frontier.

FIGURE 4 · REPORT REC. 3

6 Institutionalise and finance SDG Centers

Put SDG Centers into local planning and budgeting through MoUs or service contracts, backed by a national financing framework — budget allocation, donor guidelines, and incentives for universities to sustain SDG units.

REPORT REC. 1 & 5

Sources. SDGs Research Dashboard, SDG Center UNPAD (Scopus, June 2026), for the coverage, leadership and concentration figures. OpenAlex and Scopus, 2015–2026, for the quality-filter figure. *SDG Center Business Model — Final Report* (2025), prepared for the Asian Development Bank, for the alignment figures.